# Direct Head-to-Head: RLSTCcode vs Q-RLSTC

## Metric Definition

Competitive Ratio (CR) measures segmentation-induced clustering quality relative to an unsegmented baseline:

$$\mathrm{CR} = \frac{\mathrm{OD}_{\mathrm{segmented}}}{\mathrm{OD}_{\mathrm{baseline}}}$$

where OD is the average over-distance (mean IED from each sub-trajectory to its assigned cluster centre). Lower CR is better: CR < 1.0 means the RL agent produces tighter clusters than the baseline; CR = 1.0 is parity; CR > 1.0 is worse.

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## All Models Ranked by Competitive Ratio

| Rank | Model | System | Params | Val CR | Optimizer | Training Data |
|---|---|---|---|---|---|---|
| 1 | RLSTCcode (best, 3k) | RLSTCcode | 514 | 0.5892 | SGD | 3,000 trajs |
| 2 | RLSTCcode (5-fold CV) | RLSTCcode | 514 | 0.7543 +/- 0.0369 | SGD | ~4,000 trajs |
| 3 | VQ-DQN (5q x 3L) | Q-RLSTC | 34 | 1.4811 | SPSA | 30 trajs |
| 4 | RLSTCcode (model-state4) | RLSTCcode | 514 | 1.5453 | SGD | unknown |
| 5 | Control C (h=32x32) | Q-RLSTC | 1,314 | 1.6390 | SPSA | 30 trajs |
| 6 | Control B (h=64) | Q-RLSTC | 514 | 5.1133 | SPSA | 30 trajs |
| 7 | Control A (linear) | Q-RLSTC | 12 | 9.2051 | SPSA | 30 trajs |

> Uncertainty note: RLSTCcode reports 5-fold CV variance. Q-RLSTC results are single-seed (seed=42). Multi-seed confidence intervals for the SPSA models are a required follow-up (see Limitations).

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## Claim A: Absolute Clustering Quality

RLSTCcode achieves the best absolute CR when given sufficient data and gradient access.

| Dimension | RLSTCcode (best) | VQ-DQN |
|---|---|---|
| Val CR | 0.5892 | 1.4811 |
| Parameters | 514 | 34 |
| Training data | 3,000 trajs | 30 trajs |
| Optimizer | SGD (backprop) | SPSA (gradient-free) |
| CR ratio | 1.0x | 2.5x worse |

| Dimension | RLSTCcode (best) | VQ-DQN |
|---|---|---|
| Data ratio | 1.0x | 100x less |

RLSTCcode wins on absolute CR (0.5892 vs 1.4811) thanks to 100x more training data and gradient-based optimization. This comparison conflates three variables (optimizer, data scale, model class) and is not a controlled experiment.

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## Claim B: Parameter-Efficient Inductive Bias under SPSA

Under identical gradient-free training conditions, the 34-parameter VQ-DQN outperforms all classical controls including one with 39x more parameters.

Controlled Conditions (identical for all four models)

| Dimension | Value |
|---|---|
| Optimizer | SPSA (gradient-free, shared implementation) |
| Training data | 30 trajectories (27 train / 3 val) |
| Epochs | 2 |
| Batch size | 32 |
| Replay buffer | 5,000 |
| Gamma | 0.9 |
| Epsilon schedule | 1.0 -> 0.1 (decay 0.99) |
| Target update | Hard copy every 10 episodes |
| Double DQN | Yes |
| Seed | 42 |

SPSA Hyperparameters (shared by all models)

| Parameter | Value | Role |
|---|---|---|
| A (stability) | 20 | Warmup period for learning rate |
| a (initial LR) | 0.12 | Learning rate scale |
| c (perturbation) | 0.08 | Finite-difference step size |
| alpha | 0.602 | LR decay exponent |
| gamma | 0.101 | Perturbation decay exponent |
| Grad clip | 10.0 | Max gradient norm |
| Momentum (m-SPSA) | Disabled for E1 | No gradient averaging |
| Tuning budget | None | Defaults from Spall (1998) |

Results

| Model | Params | Val CR | CUT% | Segments | Wall Time |
|---|---|---|---|---|---|
| VQ-DQN (5q x 3L) | 34 | 1.4811 | 33.2% | 465 | 987s |
| Control C (h=32x32) | 1,314 | 1.6390 | 3.9% | 57 | 10s |
| Control B (h=64) | 514 | 5.1133 | 0.2% | 6 | 9s |
| Control A (linear) | 12 | 9.2051 | 0.0% | 3 | 9s |

VQ-DQN achieves 9.6% lower CR with 39x fewer parameters than the best classical control.

CUT% Interpretation

The VQ-DQN's 33.2% cut rate produces 465 segments from 30 trajectories (avg ~15.5 segments per trajectory). This is not degenerate over-segmentation: the D1 random-baseline experiment shows that random cutting at p=0.5 achieves CR=1.4311 with 453 segments (comparable segment count, slightly better CR). The VQ-DQN's learned policy achieves a similar segment count through selective, reward-driven cuts rather than uniform random ones.

For context, min-segment-length L_MIN=3 prevents trivially short segments, and the CUT_PENALTY=0.12 in the reward function explicitly penalises excessive cutting.

Same-Architecture Comparison (Dense 5->64->2, 514 params)

| Model | Val CR | Optimizer | Data | Result |
|---|---|---|---|---|
| RLSTCcode | 0.5892 | SGD | 3,000 | Learns effective policy |
| Q-RLSTC Control B | 5.1133 | SPSA | 30 | Fails to learn cutting |

> The identical 514-param MLP degrades 8.7x when switching from SGD+3k data to SPSA+30 data. This confirms that the gap between Claim A and Claim B is driven by optimizer and data scale, not architecture.

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## Takeaway

With the same gradient-free optimizer and the same 30-trajectory training regime, a 34-parameter VQ-DQN matches or beats substantially larger classical controls, suggesting parameter-efficient inductive bias in the small-data SPSA setting — while classical SGD with sufficient data still wins on absolute CR.

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## Limitations

1. Multi-seed uncertainty: SPSA results are from a single seed (42). Confidence intervals across 5+ seeds are needed to confirm statistical significance of Claim B.
2. Data-scale confound: The 100x data gap between Claim A and Claim B prevents fair absolute comparison. Running RLSTCcode at 30 trajectories (or Q-RLSTC at 3,000) would isolate the model-class variable.
3. SPSA tuning: All models share default SPSA hyperparameters (Spall 1998). It is possible that architecture-specific tuning of (a, c, A) could improve classical control performance under SPSA.
4. Epoch count: 2 epochs is minimal. Extended training may narrow or widen the quantum-classical gap.

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## Plots

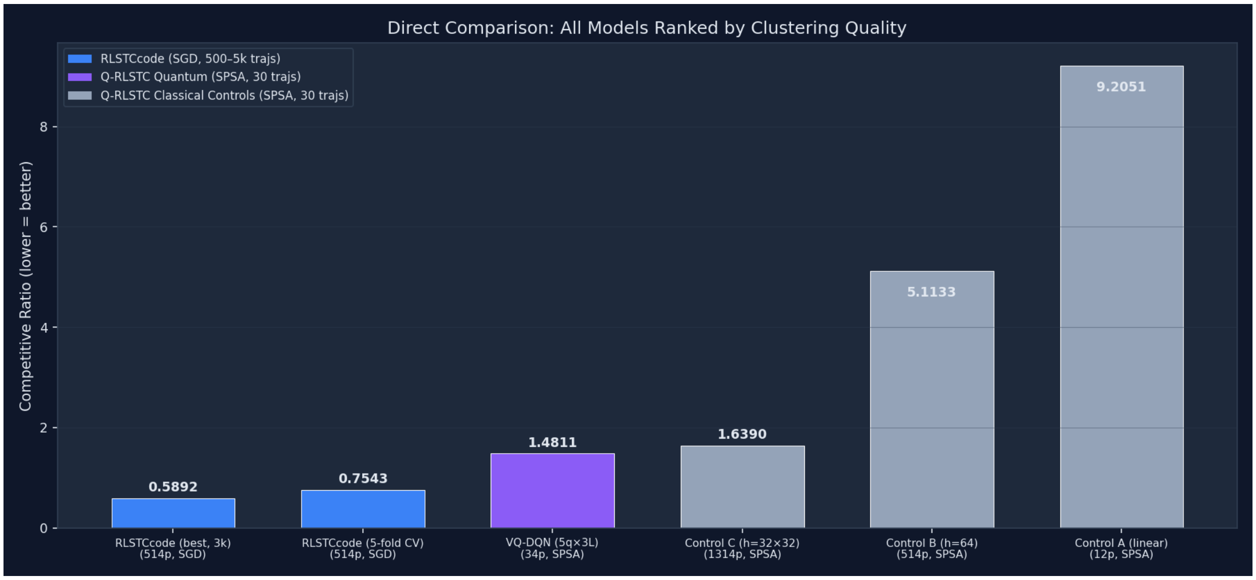


Figure 1: Unified Comparison

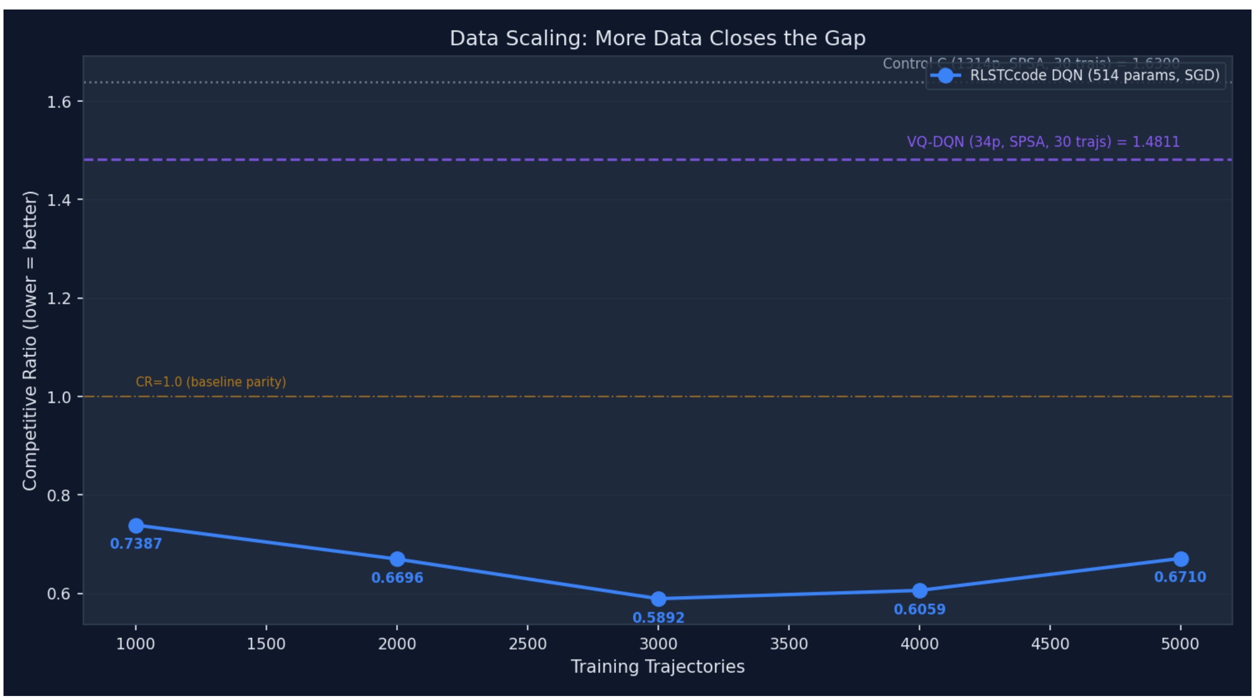


Figure 2: Data Scaling

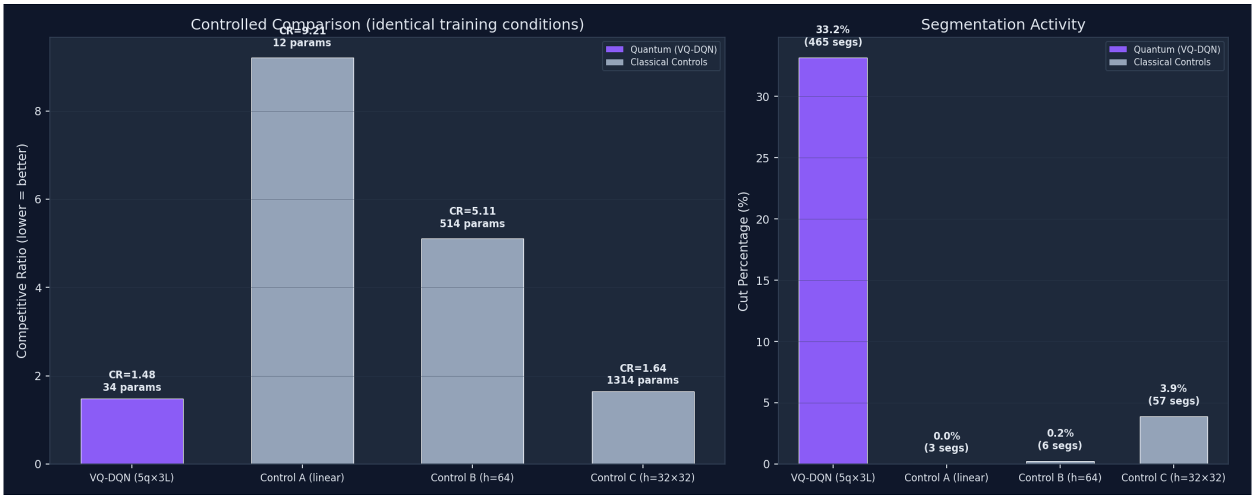


Figure 3: Controlled Comparison


Figure 4: Parameter Efficiency


Figure 5: Summary Dashboard